VisionGuard AI: A Multimodal Framework for Intelligent Patient Monitoring

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ABSTRACT ***Patients in ICUs and observation units continue to be monitored primarily through physiological sensors and periodic nurse rounds. While sensors excel at detecting abnormalities in vital signs, they provide no insight into patient behavior. Falls, unconsciousness, pain and discomfort are frequently among the initial signs of impending issues, yet none of these are evident on a pulse oximeter. VisionGuard AI is a multi-modal monitoring system that integrates video from an ESP32-CAM with heart rate, SpO_2 and body temperature readings from IoT sensors. A vision module analyzes video footage for unusual behavior, falls and prolonged inactivity. A sensor module employs a trained deep-learning model to detect early signs of deterioration in the vital signs. The outputs from both modules are merged in a fusion layer to assign patients to one of four risk categories (Normal, Moderate, High, Critical), and an Explainable AI (XAI) component displays the evidence that supports each alert. Alerts are transmitted via a buzzer and a real-time dashboard. During prototype testing, using a trained physiological model and an OpenCV-based visual reference running in aStreamlit dashboard, the system successfully predicted patient deterioration before predefined thresholds were exceeded, identified bed falls and prolonged inactivity, and maintained stable telemetry in real-time. These preliminary findings indicate that integrating visual and sensor data can enhance patient monitoring speed and reliability.***

INDEX TERMS ***Multimodal Fusion, Patient Monitoring, Vision Transformer, IoT Sensors, Explainable AI, Fall Detection, Healthcare AI.***

I. INTRODUCTION

In the majority of ICUs and observation units, monitoring is achieved using two primary tools: sensors attached to the patient, and nurses conducting rounds. Sensors capture measurable changes effectively, such as decreases in heart rate or oxygen saturation levels. However, sensors struggle to illustrate what is physically occurring with the patient. A fall from the bed, fainting spells, noticeable pain or extended periods of motionless behavior can occur between rounds, and none will necessarily cause immediate changes in sensor

readings. By the time a clinician observes, precious minutes may have passed.

VisionGuard AI views the camera and physiological sensors as two distinct forms of evidence that must be analyzed collectively. An ESP32-CAM continuously monitors the patient, while an ESP32-based sensor node sends heart rate, SpO_2 and temperature readings via Wi-Fi. A vision model evaluates the video feed for abnormal postures, sudden falls and prolonged inactivity, whereas a sensor model analyzes the vital sign stream for early indicators of collapse. Due to the integration of both streams, VisionGuard AI avoids waiting for a single threshold to be met before evaluating patient behavior and vital signs. Like a nurse at the bedside, VisionGuard AI assesses patient behavior and vital signs concurrently. If the combined assessment indicates a risk to patient safety, VisionGuard AI assigns a level of severity to the situation, issues an alert and provides an explanation outlining which visual or physiological evidence supported the alert.

The same concept applies beyond the ICU setting, in general wards, step-down units, elderly care facilities and home rehabilitation following surgery where continuous one-on-one nursing is typically unavailable. VisionGuard AI is not intended to replace medical personnel. Instead, it is designed to function as an additional pair of eyes that remain vigilant at all times and direct medical staff to the moments requiring their intervention.

II. LITERATURE REVIEW

The field of AI-based patient monitoring has evolved from studies focused on either vision-only or sensor-only approaches to investigations examining multi-modal strategies. Each area of research contributes positively to understanding how to monitor patients safely; however, each also has limitations that VisionGuard AI seeks to address.

Alzahrani and Alghamdi [1] combined a Vision Transformer with a Bidirectional GRU (BiGRU) for fall recognition and reported an impressive 99.67% accuracy rate on benchmark video data. Their findings demonstrate that transformer models capture the spatial and temporal nuances required to distinguish falls from typical movements better than conventional convolutional architectures, and their study is the foundational study for our vision module. Patel and Biradar [2] utilized an older yet widely employed pipeline consisting of Histogram of Oriented Gradients (HOG), optical flow features and a Support Vector Machine to classify patient behavior; demonstrating that inexpensive classical algorithms can still provide a viable baseline. Wang et al. [3] examined non-contact video-based vital sign monitoring in hospitals, estimating physiological parameters from skin color changes and subtle movements captured by standard cameras. Rani et al. [4] investigated IoT and AI for predictive healthcare telemetry and demonstrated that machine learning models utilizing networked sensor data can anticipate patient deterioration rather than simply reacting to threshold violations.

Together these studies point in a similar direction. Most existing systems rely on a single modality; vision methods focus on one type of behavior (e.g., falls), sensor methods focus solely on physiological trends, and rarely are the two modalities combined. Additionally, most systems do a poor job explaining their predictions; thus, when clinicians receive an alert from one of these systems they have little idea why it was triggered.

A gap exists for a multi-modal, risk-based framework that integrates visual and physiological evidence, grades risk across multiple levels (rather than providing a simple yes/no alarm), and explains its reasoning. VisionGuard AI is being developed to fill that gap.

III. PROBLEM STATEMENT

In ICUs and observation areas delays of even minutes in identifying patient deterioration can lead to severe consequences. Traditional monitoring systems rely on medical sensors and intermittent checks by nurses; however, some abnormal conditions fail to register with either method. Unusual movement patterns, falls, unconsciousness states, pain levels and states of distress may not produce any measurable changes in sensor data because sensors measure physiological function and cannot interpret behavioral responses. Conversely, a camera alone cannot identify internal physiological collapses that exhibit no outward manifestations. Therefore, a system incorporating both visual and sensor-based information is required so that abnormal conditions are less likely to be overlooked and an early warning is issued immediately upon manifestation.

IV. OBJECTIVE

The primary goal of VisionGuard AI is to integrate camera-based visual data and sensor-based physiological data into a unified and explainable patient monitoring system. VisionGuard AI has five distinct goals:

Visual Monitoring – To identify abnormal patient activity from real-time images and videos.

Physiological Monitoring – To analyze vital-sign data for abnormal physiological conditions.

Multimodal Fusion – To combine the visual and sensor outputs into an overall assessment of patient risk.

Emergency Alert – To issue a graded alert when risky conditions are detected.

Explainable AI – To explain why an alert was issued so that clinicians can trust it.

V. METHODOLOGY

VisionGuard AI utilizes a pipeline that extends from acquiring data to generating an explainable alert for clinicians. Figure 1 illustrates the complete workflow. Camera-based visual data and sensor-based physiological data are collected simultaneously and pre-processed. The visual stream then enters a vision model while the physiological stream enters a sensor model. The outputs from both models are fused together and subsequently entered into a risk-detection stage. If an anomaly is identified an alert is produced; if not then monitoring continues. Regardless of whether an anomaly is identified or not, the decision is passed through an Explainable AI stage prior to repeating the cycle; consequently every outcome

includes an explanation.

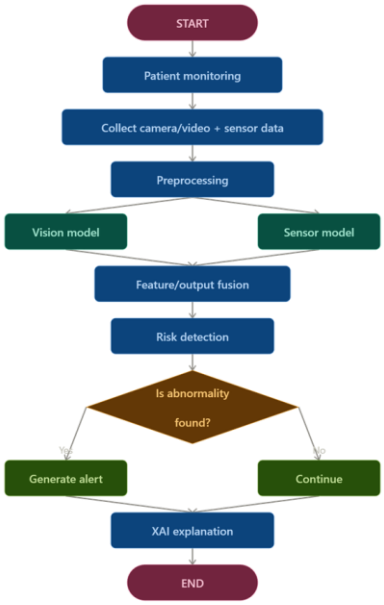


Fig. 1. Flow Diagram Illustrating Overall VisionGuard AI Monitoring Pipeline

Figure 2 illustrates the system architecture supporting this workflow. At the patient end there exist two parallel pathways for data acquisition. The camera stream enters vision pre-processing and subsequently enters a deep-learning vision model. The sensor stream enters sensor pre-processing and subsequently enters either a machine-learning or deep-learning model. Both outputs converge at the multimodal fusion layer where a single risk prediction is generated. That prediction and its corresponding XAI output enter the alert module which interfaces with the clinical response loop comprising clinician review and ICU care actions. It is this closed loop extending from raw sensing to an alert a clinician can respond to that permits VisionGuard AI to operate as an automated extension of ward-level oversight.

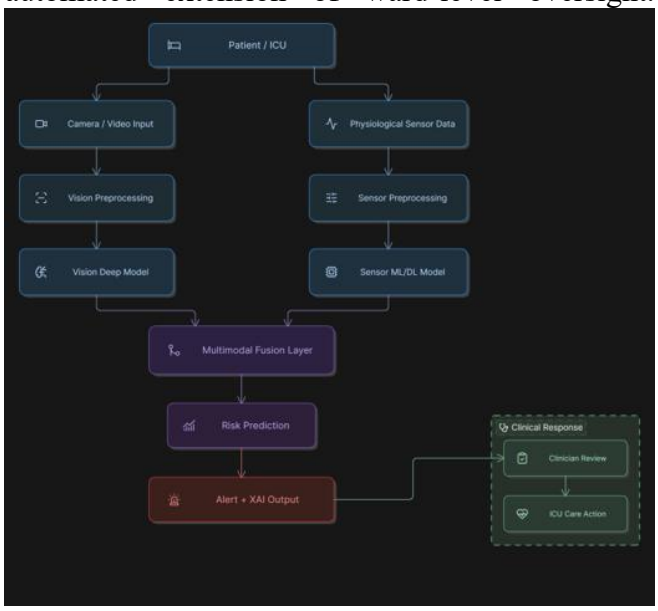


Fig. 2. System Architecture Supporting Proposed VisionGuard AI Framework

VisionGuard AI consists of three modules. Module 1 (Vision-Based Detection) extracts features from patient images/videos; identifies anomalous behavior; and categorizes the scene as either normal or anomalous. Module 2 (Sensor-Based Detection) processes physiological parameters; identifies anomalous patterns; and predicts patient risk based solely on vital signs. Module 3 (Multimodal Fusion and Explainable AI) integrates predictions from both modules to improve decision-making accuracy; produces emergency alerts; and explains its output using XAI techniques such as Grad-CAM or SHAP.

VI. IMPLEMENTATION

A. Environment Setup

The prototype was developed using Python within Visual Studio Code. TensorFlow/Keras and PyTorch were used for model development; OpenCV for visual processing; NumPy/Pandas/Scikit-learn/Matplotlib for data

management/evaluation/plotting; Grad-CAM/SHAP for explainability; Hardware-wise an ESP32-CAM captures patient video; an ESP32 development board sends sensor data via Wi-Fi; a MAX30102 sensor measures heart rate/SpO_2; and an MLX90614 infrared sensor measures body temperature. A breadboard with jumper wires connected components during prototype development.

B. Dataset Preparation

Physiological data consists of sequential multi-variable records of heart rate/SpO_2/body temperature; labeled as stable/collapsing trajectories for training the sensor model. For the visual branch; UR Fall/ADL (Activities of Daily Living) datasets have been identified as training data for upgrading Vision Transformer capabilities. The current prototype utilizes a labeled set of short clips depicting normal movement/bed falls/post-fall immobility for validation of baseline functionality.

C. Model Design

The physiological branch is a PyTorch multi-layer perceptron (MLP) stored as vital_sign_model.pth. It processes sequential vital-sign data; flags multi-variable collapse trajectories early; rather than awaiting establishment of a single threshold. The visual branch is currently an OpenCV baseline built upon temporal frame differencing/centroid tracking. On top sits state-latching logic: once a fall/sudden impact is detected; status remains critical until clearing; thus preventing reversion to normal status during an actual emergency. This baseline is intended for replacement by Vision Transformer/3D-CNN pose-estimation model utilizing approaches similar to those employed by Alzahrani/Alghamdi [1].

D. Training/Validation

The MLP was trained on labeled sequential vital-sign data with a held-out validation set utilized to confirm ability to differentiate stable trajectories from those approaching collapse prior to static thresholds triggering alarms. The visual baseline was tested on recorded clips depicting normal activity/bed falls/rapid impacts/prolonged immobility.

E. Evaluation

Both branches are evaluated based on accuracy/precision/recall/F1-score. Recall receives significant emphasis because false negatives (missed detections) represent the most costly errors in safety-critical applications.

F. Deployment/Inference

Trained physiological model/visual baseline were incorporated into a Streamlit web application providing real-time inference on incoming data. The dashboard displays live telemetry/rolling line charts depicting vitals/side-by-side comparison between conventional threshold-based monitoring/deep-learning early-warning outputs. Clinicians can monitor both modalities and their respective alerts from one screen.

VII. RESULTS & DISCUSSION

A. Performance of Physiological Model

PyTorch MLP (vital_sign_model.pth) operated successfully for real-time inference on sequential test data. It generated early warnings for multi-

variable collapse trajectories prior to static thresholds being exceeded. This represents a substantial improvement over conventional monitors which only alarm once a hard limit such as heart rate >120 BPM/SpO_2 <90% is exceeded.

B. Performance of Visual Monitoring Branch

OpenCV baseline utilizing temporal frame differencing/centroid tracking/state latching identified bed falls/rapid impacts/critical post-fall immobility in test recordings. Figure 3 illustrates dashboard flagging critical fall event/displaying "PATIENT DOWN ON FLOOR" alert/latching emergency status until cleared.

Fig. 3. VisionGuard AI Dashboard Flagging Critical Fall Event



Fig. 3. VisionGuard AI dashboard detecting a critical fall event in real time.

C. Integration into Dashboard

Fully interactive Streamlit application provides live telemetry/rolling line charts/detection comparison side-by-side. Figure 4 illustrates patient stability: heart rate/SpO_2/temperature are within normal range/threshold-based monitor/deep-learning engine agree/engine reports high confidence level regarding assessment.



Fig. 4. Real-time vitals telemetry and detection-engine comparison on the VisionGuard AI dashboard.

D. Overall Analysis

Across both branches prototype supports central idea behind VisionGuard AI. Physiological model can warn earlier than static thresholds/visual baseline can independently verify events (such as falls) sensor cannot perceive. Results from both branches complement each other rather than overlap; which is behavior planned fusion layer is intended to capitalize upon. Combining both signals should provide more confident/more explainable risk assessment than either modality individually. Note that fusion layer itself is yet to be implemented/evaluated on real hardware.

VIII. APPLICATIONS

A. ICU/Critical Care Monitoring

Provides continuous/automated observation of ICU patients/enhances nursingrounds/physiological sensors with visual behavior monitoring.

B. Emergency Fall Detection/Trauma Response

Automatically flags falls/sudden impacts/post-fall immobility; enabling wards/emergency

departments/rehabilitation units to respond more quickly.

C. Elderly/Home-Based Care

Supports remote monitoring of elderly/post-operative patients at home/alerts caregivers/family members to falls/physiological distress.

D. Telemedicine/Remote Healthcare

Extends continuous monitoring capabilities to rural clinics/step-down facilities incapable of providing continuous one-on-one nursing care.

E. Clinical Decision Support/Second Opinion

Provides clinicians with graded/explainable risk assessment alongside their own judgment; therebyminimizing likelihood that subtle deterioration will be overlooked during busy shifts.

F. Nurse Station Alert Integration

Can interface with SMS/push-notification services; enabling critical alerts to be transmitted directly to nurse station pagers/mobile devices.

G. Hospital Resource Optimization

Facilitates allocation of limited nursing time towards patients categorized as moderate/high/critical risk.

IX. FUTURE SCOPE

Phase 2 Hardware Deployment – Flash physical ESP32 boards/MAX30102 pulse oximeters/MLX90614 temperature sensors; thereby enabling streaming live data into dashboard.

Advanced Vision Models – Replace OpenCV baseline with Vision Transformer (ViT)/3D-CNN pose-estimation model trained on UR Fall/ADL datasets.

Clinical Notification Webhooks – Add SMS/push-notification APIs enabling transmission of emergency alerts to nurse station pagers.

Federated/Multi-Site Learning – Train models across hospitals without exchanging raw patient data; improving generalizability/preserving privacy.

Regulatory Validation – Conduct prospective clinical trials/pursue regulatory approval for safe deployment in real-world environments.

X. CONCLUSION

VisionGuard AI illustrates that combining visual observation/physiological sensing can bridge a significant gap in patient monitoring. Sensors cannot perceive a fall/camera cannot read a vital sign; but together they offer a more comprehensive description of patient condition. In our prototype; physiological model provided early warnings prior to static thresholds being exceeded/visual baseline detected falls/immobility in real-time/Streamlit dashboard presented both streams in a single interface for clinicians. Next steps include implementing Vision Transformer for visual branch/deploying functioning fusion layer/deploying on real hardware. The ultimate goal is dependable/explainable second observer at

bedside assisting nurses/doctors identify problems earlier/preserve patient safety.

XI. REFERENCES

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